# Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

## *Insulation systems*

from

## *KIMFOR*

| | |
|---|---|
| Programme: | The International EPD System, www.environdec.com |
| Programme operator: | EPD International AB |
| Licensee: | In the case of EPDs registered through a regional or national licensee |
| Type of EPD: | Choose from Table 1 of GPI V5.0.1 |
| EPD registration number: | EPD-IES-00XXXXX |
| Version date: | 20YY-MM-DD |
| Validity date: | 20YY-MM-DD |

*An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com*

# GENERAL INFORMATION

| Programme Information | |
|---|---|
| **Programme:** | The International EPD® System |
| **Address:** | EPD International AB<br>Box 210 60<br>SE-100 31 Stockholm<br>Sweden |
| **Website:** | www.environdec.com |
| **E-mail:** | support@environdec.com |

| Product Category Rules (PCR) |
|---|
| **CEN standard EN 15804 serves as the Core Product Category Rules (PCR)** |
| **Product Category Rules (PCR):** ***PCR 2019:14 Construction products (EN 15804+A2) (2.0.1)*** |
| **PCR review was conducted by:** ***The Technical Committee of the International EPD® System. See www.environdec.com/TC for a list of members. Review chair: Claudia A. Peña, University of Concepción, Chile. The review panel may be contacted via the Secretariat www.environdec.com/contact*** |

| Third-party Verification |
|---|
| Independent third-party verification of the declaration and data, according to ISO 14025:2006, via:<br><br>☒ **Individual EPD verification without a pre-verified LCA/EPD tool**<br>Third-party verifier: xxx<br>Approved by: International EPD System |
| Procedure for follow-up of data during EPD validity involves third party verifier:<br><br>☐ Yes ☒ No |

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same first-digit version number) or be based on fully aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant); apply identical impact assessment methods (including the same version of characterisation factors); and be valid at the time of comparison.
For further information about comparability, see EN 15804 and ISO 14025.

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# INFORMATION ABOUT EPD OWNER

Owner of the EPD: Kimfor Yapi Kimyasallari San.Ve Tic.Ltd.Sti
Address: Yunuseli Mh. Akyıldız Cd. No: 3 Osmangazi / Bursa / Turkey
Contact: Ismail Kuscuoglu and Korra Hubnext; +90 850 303 31 11; info@kimfor.com.tr
Address and contact information of the LCA practitioner commissioned by the EPD owner, if applicable: Quality Net Srl
Description of the organisation: It is a leading company in the development, production, and introduction of new generation ecological insulation systems to the international market. The ecological composite insulation products developed under the brand name THERMOLOCK, offer a high level of safety and comfort by insulating living spaces against heat, sound, water, radiation, and fire. They also offer added value to the economy by being manufactured using local raw materials of Turkey, where the use of imported materials predominates. The products are exported to many different countries worldwide, and the aim is to establish a place for finished products on the international market, rather than relying on the export model of raw materials and semi-finished products.
Product-related or management system-related certifications: Kimfor Insulation Construction Chemicals Ltd. operates an integrated Quality and Evironmentral Management system in accordance with ISO 9001:2015 and ISO 14001:2015.

# PRODUCT INFORMATION

Product name: Thermolock Thermal Insulation Plaster
Product identification: Decorative surface coating

UN CPC code: 37410 - Plasters
Product description: Plaster for interior use, for exterior use, for roofs, base product. These products are characterized by unique properties thanks to the use of raw materials extracted in Turkey, which ensure water resistance, fire resistance, and high durability under extreme temperatures.
Name and location of production site(s): Fındıklı Road No: 110, Orhangazi/Bursa, Turkey
https://thermolock.com/en

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# CONTENT DECLARATION

- The mass (weight) of one unit of a product, as purchased or per declared unit: 1 kg.
- Content of the product in the form of a list of materials and substances, and their mass:

| Product content | Mass, kg | Post-consumer recycled material, mass-% of product | Biogenic material, mass-% of product | Biogenic material, kg C/product or declared unit |
|---|---|---|---|---|
| Cement | 0,20 | 0 | 0 | 0 |
| Mineral fillers (calcite) | 0,35 | 0 | 0 | 0 |
| Mineral fillers (quartz) | 0,15 | 0 | 0 | 0 |
| Mineral fillers (perlite) | 0,25 | 0 | 0 | 0 |
| Polymer DLP 2140 | 0,02 | 0 | 0 | 0 |
| Polymer Arbocell PWC 500 | 0,01 | 0 | 0 | 0 |
| Additives (Zinc Stereat) | 0,01 | 0 | 0 | 0 |
| Calcium stearate | 0,01 | 0 | 0 | 0 |
| TOTAL | 1 | 0 | 0 | 0 |

- The mass and the content of distribution and/or consumer packaging:

| Packaging materials | Mass, kg | Mass-% (versus the product) | Biogenic material, kg C/product or declared unit |
|---|---|---|---|
| Kraft | 0,0064 | 0,0064 | 0 |
| Wooden | 0,027 | 0,027 | 0 |
| Nylon | 0,00093 | 0,00093 | 0 |

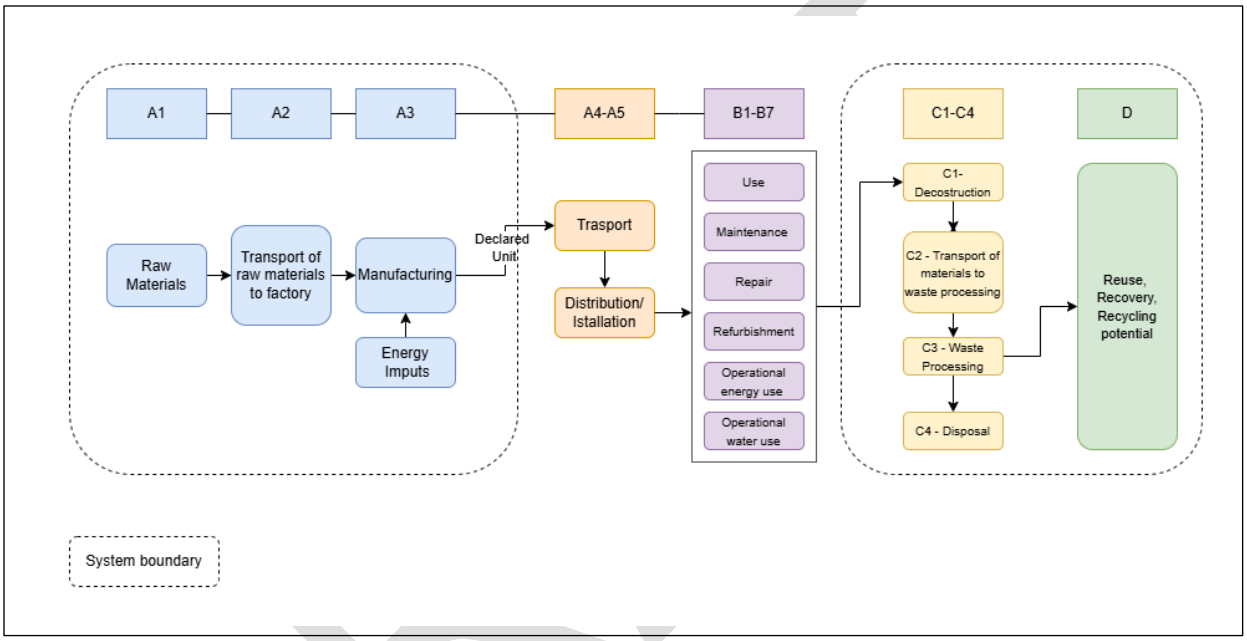
- Information on the environmental and hazardous/toxic properties of a substances contained in the product: Not applicable*
- Other information on substances with hazardous and toxic properties: Not applicable*
- The declared share of biogenic/recycled materials: 0

*No substances listed under the SVHC (Substances of Very High Concern) list, as defined by the REACH Regulation, have been identified as being present in the finished product.

# LCA INFORMATION

Functional unit 1 kg
Time representativeness: 2024
Geographical scope: Turkey (A1-A3, C1-C4, D)
Database(s) and LCA software used: Ecoinvent 3.10 SimaPro 10.3.0.1
Description of system boundaries: Cradle to gate with modules C1–C4 and module D (A1–A3 + C + D);
Process flow diagram:

*Figure 1: Process flow diagram*




The operational processes can be summarized as follows:

More information:

Data Quality summary: the data quality was assessed in accordance with the Commission Recommendation 2013/179/EU (PEF) and UNI EN 15804:2019 (Annex E). For each dataset, three scores were assigned related to temporal representativeness (TiR), technical representativeness (TeR), and geographical representativeness (GR), following the criteria set out in Table 5 of the PEF Recommendation. The overall Data Quality Rating (DQR) was then calculated as the arithmetic mean of these three values: DQR = (TiR + TeR + GR) / 3. Based on the obtained value, datasets were classified into quality levels ranging from “excellent” to “poor,” with only those achieving at least a “good quality” level being considered acceptable. Overall, the data quality was considered “Good Quality” and no data were excluded from the analysis in the LCA report. A summary is reported in the table below.

| Process name | Source type | Source | Relevance year | Data category | Share of data, of GWP-GHG results for A1-A3 | Relevance | Temporal representativeness | Technical representativeness | Geographical representativeness | DQR | Quality rating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Cement | Company data | Ecoinvent 3.10 | 2024 | secondary | 14% | 14% for GWP GHG; for others from 2% (Land use) to 31% (Climate change - Fossil) | 2 | 2 | 2 | 2 | Very good quality |
| Calcite | Company data | Ecoinvent 3.10 | 2024 | secondary | 11% | 11% for GWP GHG; for others from 2% (Climate change - Biogenic) to 25% (Resource use, minerals and metals) | 2 | 2 | 2 | 2 | Very good quality |
| Perlite | Company data | Ecoinvent 3.10 | 2024 | secondary | 9% | 9% for GWP GHG; for others from 2% (Land use) to 24% (Eutrophication, terrestrial) | 2 | 2 | 2 | 2 | Very good quality |
| DLP 2140 | Company data | Ecoinvent 3.10 | 2024 | secondary | 16% | 16% for GWP GHG; for others from 2% (Land use) to 36% (Ozone depletion) | 2 | 2 | 2 | 2 | Very good quality |
| Zinc Stereat | Company data | Ecoinvent 3.10 | 2024 | secondary | 10% | 10% for GWP GHG; for others from 0,56% (Land use) to 53% (Ecotoxicity, freshwater - part 2) | 2 | 2 | 2 | 2 | Very good quality |
| Packaging | Company data | Ecoinvent 3.10 | 2024 | secondary | 8% | 8% for GWP GHG; for others from 0,86% (Resource use, minerals and metals) to 58% (Climate change - Biogenic) | 2 | 2 | 2 | 2 | Very good quality |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| Upstream raw material transport | Company data | Ecoinvent 3.10 | 2024 | primary | 12% | 12% for GWP GHG; for others from 1% (Climate change - Biogenic) to 22% (Particulate matter) | 2 | 2 | 2 | 2 | Very good quality |
| | | | | | | | | | | 2,08 | Good quality |

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Modules declared, geographical scope, share of primary data (in GWP-GHG results) and data variation (in GWP-GHG results):

| | Product stage | | | Distribution/ installation stage | | Use stage | | | | | | | End-of-life stage | | | | Beyond product life cycle |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Raw material supply | Transport | Manufacturing | Transport | Construction installation | Use | Maintenance | Repair | Replacement | Refurbishment | Operational energy use | Operational water use | De-construction demolition | Transport | Waste processing | Disposal | Reuse-Recovery-Recycling-potential |
| **Module** | **A1** | **A2** | **A3** | **A4** | **A5** | **B1** | **B2** | **B3** | **B4** | **B5** | **B6** | **B7** | **C1** | **C2** | **C3** | **C4** | **D** |
| Modules declared | x | x | x | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND |
| Geography | TR | TR | TR | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Share of primary data | 16,66% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Variation – products | 0% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Variation – sites | 0% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |

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# ENVIRONMENTAL PERFORMANCE

## LCA results of the product(s) - main environmental performance results

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

### Mandatory impact category indicators according to EN 15804

| Results per functional or declared unit | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A4** | **A5** | **B1** | **B2** | **B3** | **B4** | **B5** | **B6** | **B7** | **C1** | **C2** | **C3** | **C4** | **D** |
| GWP-total | kg $CO_2$ eq. | 4,50 E-01 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,65 E-02 | 0,00 E+00 | 1,99 E-02 | 0,00 E+00 |
| GWP-fossil | kg $CO_2$ eq. | 4,08 E-01 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,64 E-02 | 0,00 E+00 | 1,96 E-02 | 0,00 E+00 |
| GWP-biogenic | kg $CO_2$ eq. | 4,18 E-02 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 8,88 E-05 | 0,00 E+00 | 2,31 E-04 | 0,00 E+00 |
| GWP-luluc | kg $CO_2$ eq. | 2,87 E-04 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 6,49 E-06 | 0,00 E+00 | 4,88 E-05 | 0,00 E+00 |
| ODP | kg CFC 11 eq. | 4,88 E-09 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 2,28 E-10 | 0,00 E+00 | 3,82 E-10 | 0,00 E+00 |
| AP | mol $H^+$ eq. | 1,97 E-03 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 5,45 E-05 | 0,00 E+00 | 1,07 E-04 | 0,00 E+00 |
| EP-freshwater | kg P eq. | 8,08 E-05 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,28 E-06 | 0,00 E+00 | 1,03 E-04 | 0,00 E+00 |
| EP-marine | kg N eq. | 5,01 E-04 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,77 E-05 | 0,00 E+00 | 4,07 E-05 | 0,00 E+00 |
| EP-terrestrial | mol N eq. | 5,48 E-03 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,92 E-04 | 0,00 E+00 | 4,41 E-04 | 0,00 E+00 |
| POCP | kg NMVOC eq. | 1,89 E-03 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 7,59 E-05 | 0,00 E+00 | 1,51 E-04 | 0,00 E+00 |
| ADP-minerals& metals* | kg Sb eq. | 5,14 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 2,29 E-01 | 0,00 E+00 | 3,31 E-01 | 0,00 E+00 |
| ADP-fossil* | MJ | 3,90 E-06 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 5,24 E-08 | 0,00 E+00 | 5,60 E-08 | 0,00 E+00 |
| WDP* | $m^3$ | 8,41 E-02 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,04 E-03 | 0,00 E+00 | 1,04 E-02 | 0,00 E+00 |
| Acronyms | GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption | | | | | | | | | | | | | | | |

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.*

## Additional mandatory and voluntary impact category indicators

| Results per functional or declared unit | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A4** | **A5** | **B1** | **B2** | **B3** | **B4** | **B5** | **B6** | **B7** | **C1** | **C2** | **C3** | **C4** | **D** |
| GWP-GHG[1] | kg $CO_2$ eq. | 4,06 E-01 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,63 E-02 | 0,00 E+00 | 1,95 E-02 | 0,00 E+00 |

## Resource use indicators

| Results per functional or declared unit | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A4** | **A5** | **B1** | **B2** | **B3** | **B4** | **B5** | **B6** | **B7** | **C1** | **C2** | **C3** | **C4** | **D** |
| PERE | MJ | 1,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 2,90 E-03 | 0,00 E+00 | 4,83 E-03 | 0,00 E+00 |
| PERM | MJ | 5,52 E-01 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| PERT | MJ | 1,56 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 2,90 E-03 | 0,00 E+00 | 4,83 E-03 | 0,00 E+00 |
| PENRE | MJ | - 1,09 E-01 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 2,77 E-04 | 0,00 E+00 | 5,07 E-03 | 0,00 E+00 |
| PENRM | MJ | 1,26 E-01 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| PENRT | MJ | 1,65 E-02 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 2,77 E-04 | 0,00 E+00 | 5,07 E-03 | 0,00 E+00 |
| SM | kg | 0,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| RSF | MJ | 0,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| NRSF | MJ | 0,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| FW | $m^3$ | 2,25 E-03 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 3,06 E-05 | 0,00 E+00 | 2,53 E-04 | 0,00 E+00 |
| Acronyms | PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water | | | | | | | | | | | | | | | |

## Waste indicators

| Results per functional or declared unit | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A4** | **A5** | **B1** | **B2** | **B3** | **B4** | **B5** | **B6** | **B7** | **C1** | **C2** | **C3** | **C4** | **D** |
| Hazardous waste disposed | kg | 5,19 E-05 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,58 E-06 | 0,00 E+00 | 7,64 E-06 | 0,00 E+00 |

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic $CO_2$ is set to zero.

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Non-hazardous waste disposed | kg | 7,62 E-02 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 1,08 E-02 | 0,00 E+00 | 1,22 E-02 | 0,00 E+00 |
| Radioactive waste disposed | kg | 3,03 E-06 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 4,62 E-08 | 0,00 E+00 | 7,26 E-08 | 0,00 E+00 |

## Output flow indicators

| Results per functional or declared unit | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Indicator** | **Unit** | **A1-A3** | **A4** | **A5** | **B1** | **B2** | **B3** | **B4** | **B5** | **B6** | **B7** | **C1** | **C2** | **C3** | **C4** | **D** |
| Components for re-use | kg | 0,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| Material for recycling | kg | 0,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| Materials for energy recovery | kg | 0,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| Exported energy, electricity | MJ | 0,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |
| Exported energy, thermal | MJ | 0,00 E+00 | ND | ND | ND | ND | ND | ND | ND | ND | ND | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 | 0,00 E+00 |

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# ABBREVIATIONS

| Abbreviation | Definition |
|---|---|
| **General Abbreviations** | |
| EN | European Norm (Standard) |
| EPD | Environmental Product Declaration |
| EF | Environmental Footprint |
| GPI | General Programme Instructions |
| ISO | International Organization for Standardization |
| LCA | Life Cycle Assessment |
| PCR | Product Category Rules |
| c-PCR | Complementary Product Category Rules |
| CEN | European Committee for Standardization |
| CLC | Co-location centre |
| CPC | Central product classification |
| GHS | Globally harmonized system of classification and labelling of chemicals |
| GRI | Global Reporting Initiative |
| **Environmental Impact Indicators (EN 15804)** | |
| GHG | Greenhouse gas |
| GWP | Global Warming Potential (kg $CO_2$ eq.) |
| GWP-fossil | Global Warming Potential from fossil sources (kg $CO_2$ eq.) |
| GWP-biogenic | Global Warming Potential from biogenic sources (kg $CO_2$ eq.) |
| GWP-luluc | Global Warming Potential from land use and land use change (kg $CO_2$ eq.) |
| GWP-total | Total Global Warming Potential (kg $CO_2$ eq.) |
| GWP-GHG | Global Warming Potential for greenhouse gases (kg $CO_2$ eq.) |
| ODP | Ozone Depletion Potential (kg CFC-11 eq.) |
| AP | Acidification Potential (mol $H^+$ eq.) |
| EP | Eutrophication Potential |
| EP-freshwater | Freshwater eutrophication potential (kg P eq.) |
| EP-marine | Marine eutrophication potential (kg N eq.) |
| EP-terrestrial | Terrestrial eutrophication potential (mol N eq.) |
| POCP | Photochemical Ozone Creation Potential (kg NMVOC eq.) |
| ADP | Abiotic Depletion Potential |
| ADP-minerals&metals | Abiotic depletion potential for non-fossil resources (kg Sb eq.) |
| ADP-fossil | Abiotic depletion potential for fossil resources (MJ) |
| WDP | Water Deprivation Potential ($m^3$) |
| **Resource Use Indicators** | |
| PERE | Use of renewable primary energy excluding renewable primary energy resources used as raw materials (MJ) |
| PERM | Use of renewable primary energy resources used as raw materials (MJ) |
| PERT | Total use of renewable primary energy resources (MJ) |
| PENRE | Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials (MJ) |
| PENRM | Use of non-renewable primary energy resources used as raw materials (MJ) |
| PENRT | Total use of non-renewable primary energy resources (MJ) |
| SM | Use of secondary material (kg) |
| RSF | Use of renewable secondary fuels (MJ) |
| NRSF | Use of non-renewable secondary fuels (MJ) |
| FW | Use of net fresh water ($m^3$) |
| Waste Indicators | |
| HW | Hazardous Waste (disposed) (kg) |
| NHW | Non-Hazardous Waste (disposed) (kg) |
| RW | Radioactive Waste (disposed) (kg) |
| **Output Flow Indicators** | |
| CFR | Components for Reuse (kg) |
| MR | Material for Recycling (kg) |
| MER | Materials for Energy Recovery (kg) |

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| | |
|---|---|
| EEE | Exported Energy, Electricity (MJ) |
| EET | Exported Energy, Thermal (MJ) |
| **Lifecycle Stages / Modules** | |
| A1 | Raw material supply |
| A2 | Transport |
| A3 | Manufacturing |
| A4 | Transport to site |
| A5 | Construction/Installation |
| B1 | Use |
| B2 | Maintenance |
| B3 | Repair |
| B4 | Replacement |
| B5 | Refurbishment |
| B6 | Operational energy use |
| B7 | Operational water use |
| C1 | Deconstruction/Demolition |
| C2 | Transport to waste processing |
| C3 | Waste processing |
| C4 | Disposal |
| D | Reuse-Recovery-Recycling potential |
| **Other Relevant Terms** | |
| SVHC | Substances of Very High Concern |
| EC No. | European Community Number |
| CAS No. | Chemical Abstracts Service Number |
| MJ | Megajoule |
| kg | Kilogram |
| $m^3$ | Cubic Meter |
| NMVOC | Non-Methane Volatile Organic Compounds |
| Sb eq. | Antimony Equivalents |
| P eq. | Phosphorus Equivalents |
| N eq. | Nitrogen Equivalents |
| CFC-11 eq. | Chlorofluorocarbon-11 Equivalents |
| $CO_2$ eq. | Carbon Dioxide Equivalents |
| kg C | Kilograms of Carbon |
| kg $CO_2$ eq. | Kilograms of Carbon Dioxide Equivalent |
| ND | Not Declared |

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# REFERENCES


a) General Programme Instructions of International EPD System. Version 5.0.1
b) PCR 2019:14. *PCR CONSTRUCTION PRODUCTS AND CONSTRUCTION SERVICES*. v 2.0.1
*c) Report LCA_Kimfor_2024_rev01*


# VERSION HISTORY

**Original Version of the EPD, 20YY-MM-DD**

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EPD
INTERNATIONAL EPD SYSTEM
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